

Multimodal VR/MR Simulation with Empathic Agents for Nursing Education

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Figure 1: Multimodal VR/MR simulation with virtual agents for clinical nursing education in my previous work [2] and on-going study.

ABSTRACT

In the field of healthcare education, including medical and nursing education, realistic and interactive simulations of various clinical scenes are required to mimic real-life situations, necessitating more than just a procedural experience. There has been a recent surge in demand for virtual/mixed reality (VR/MR) due to its advantages in providing diverse simulation environments. However, several challenges still remain to be addressed to advance to the next level of VR/MR-based clinical simulation. In this paper, I introduce the research that I have conducted on creating immersive clinical nursing scenes in VR/MR-based learning and training simulation systems, focusing on key elements such as multi-modal interaction, virtual agents, and learner performance assessment. Furthermore, I will discuss future research aimed at fostering collaboration in various healthcare fields, including nursing, and developing VR/MR simulation systems that enable learners to experience and learn from personalized clinical situations. These topics and questions will be further discussed at the IEEE VR 2024 Doctoral Consortium.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies; Applied computing—Education—Interactive learning environments;

1 INTRODUCTION

In clinical nursing education and training, simulation plays a crucial role in enabling learners to accumulate experiences that closely resemble real-life situations within a safe learning environment. It enhances learners' skills in clinical procedures, decision-making, and teamwork. Moreover, there is ample potential to enhance learn-

ers' educational outcomes by presenting various technologies and research in virtual/mixed reality (VR/MR) environments tailored to these learning objectives. I have conducted research and development centered around three key components for clinical simulation in VR/MR: (1) multi-modal interaction, (2) virtual agents, and (3) learner performance assessment.

Firstly, **multi-modal input/output**, encompassing gestures and speech, serves as a means for learners to communicate with virtual nursing tools and agents (such as virtual patients or healthcare professionals) [4]. It is essential in the process of recognizing various symptoms or signals, as well as engaging in communication with patients, their families, or the medical team, just as in a real clinical setting [3]. To facilitate learners' immersion and perception of similarity to real situations during simulations, research is required on how to use various modalities effectively.

Secondly, **virtual agents** are a key focus of interaction. They allow learners to experience clinical scenarios while interacting with virtual agents that play the roles of patients, family members, and other healthcare professionals, similar to real clinical environments [1]. This process aids learners in developing communication skills, critical thinking, and decision-making abilities, making it effective in nurturing the ability to interact with a variety of subjects in clinical settings. Consequently, it is necessary to create virtual agents capable of verbal and non-verbal communication to address different physical, psychological, and emotional states, according to their respective roles.

Finally, **learner performance assessment methods** are valuable for learners to reflect on their experiences within the simulation education program. These methods help learners identify their strengths and areas needing improvement [5]. Specifically, learners can monitor their nursing interventions, enabling educators to adjust or plan the content and methods of education accordingly. It is essential to align the simulation elements (such as tasks, procedures, and actions) performed by learners in VR/MR with the assessment criteria of educators based on the educational/training objectives.

Driven by the significance mentioned above, I have conducted research to address the following research questions. I will discuss what is needed or can be improved for my next research.

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- RQ1.** How can the multi-modal input/output be utilized to achieve realistic interaction in clinical nursing training/education, or more broadly in healthcare simulations?
- RQ2.** How can the virtual agent in the nursing training/education simulation convey emotional expressions and empathy, and enhance users' communication skills?
- RQ3.** How can we develop more effective performance evaluation methods to improve nursing students/trainees' learning efficacy and experience during clinical simulations?

2 RESEARCH PROGRESS

While focusing on the aforementioned three components, I have developed VR/MR-based clinical nursing education/training simulations and evaluated them for learning experience.

In my previous publication at IEEE ISMAR 2022 [2], I proposed a framework for evaluating pediatric nursing interventions performed by learners using multi-modal input in immersive VR and creating virtual agents that respond emotionally based on the evaluation results (see Figure 1). I developed a module to assess the impact of users' *multi-modal inputs (hand gestures/speech)* on virtual patients, considering both discrete nursing interventions (e.g., a single word or gesture) and cumulative nursing interventions (multiple actions) performed by the users. In clinical nursing, the interventions are not isolated actions (e.g., a single word) but rather accumulative actions. Therefore, various judgments were made regarding users' multi-modal inputs. In this process, both *domain-based interaction behavior evaluation* for individual actions and *machine learning-based detection* of the cumulative nursing interventions were used to assess the actions that the user was performing at the time. In the field of pediatric nursing, it is often the case that a patient's parent/caregiver conveys the patient's wishes or reacts more sensitively to changes in the patient's condition. Therefore, appropriately addressing such situations in the clinical nursing environment is particularly important. In the conducted study, I designed a system where the emotional responses (based on the methods in the psychological studies) of *virtual parents/caregivers* are updated in real-time, depending on the nursing interventions.

Using the proposed framework, I developed a pediatric nursing scenario in collaboration with a nursing faculty at my university for clinical nursing education simulations involving pediatric patients with pneumonia at the time of discharge. In the actual user study, I conducted comparative experiments between scenarios where a virtual parent (mother) exhibited dynamic emotional responses based on users' (nursing trainees') multi-modal nursing interventions and scenarios where there were no emotional responses (statically neutral emotion) (see Figure 1). The experiment involved 9 registered nurses from South Korea with nursing qualifications and clinical nursing experience. Participants performed seven nursing interventions (e.g., patient assessment, explanation of discharge medications, fall prevention, etc.) in the two agent conditions. In qualitative comparison, 'learning efficacy' and 'usefulness' among the 4 categories (immersion, realism, learning efficacy, and usefulness) are higher with the emotional agent. In addition, the accuracy of predicting accumulated nursing interventions for performance evaluation was assessed, yielding positive results with high accuracy.

Recent research proposes an effective clinical nursing education simulation framework in Augmented Reality (AR), utilizing a nursing preceptor (a senior nurse in the hospital) agent to provide guidance and knowledge for a virtual patient's multi-modal symptoms. Specifically, among various methods to make realistic clinical nursing simulations, this framework enables learners to recognize subtle *multi-modal symptoms/signs* (including smell, temperature, and respiration with physical devices) of patients and learn appropriate nursing interventions. In the framework, *learners' nursing performance* is also assessed using LSTM (Long short-term memory) network, and detailed explanations by the *virtual nursing preceptor*



Figure 2: Example scenes of the user's multi-modal interaction with the virtual patient (left: VR, right: MR).

provide clinical nursing education similar to that in actual hospitals, offering learners a personalized learning experience. Using the proposed framework, AR-based nursing intervention systems and scenarios, including equipment for multi-modal symptoms/signs, were developed, and the learning effectiveness was verified quantitatively and qualitatively with professional nurses.

3 PLANNED RESEARCH

In future research, I will continue to focus on studies involving emotionally expressive (empathic) virtual agents, particularly in clinical nursing and healthcare contexts with user's multi-modal emotional cues. I studied the effects of embodied virtual agents on user immersion and learning outcomes in previous research (Figure 2). In healthcare simulations, particularly in the dynamic fields of nursing and medicine where diverse clinical training is essential, it becomes imperative for virtual agents—whether they represent patients or medical staff—to exhibit a high degree of freedom in responding to users' emotional inputs. With the recent advancements in artificial intelligence and various technologies, environments have been developed that allow for real-time simulations in immersive virtual/mixed environments with interactions similar to those with real people. Throughout the simulation process, the goal is not only to assess the accuracy of user actions aligned with learning objectives, as demonstrated through behaviors and speech but also to discern the user's emotional state. This understanding of the user's emotions facilitates the delivery of personalized education and training simulations, thereby enhancing the overall effectiveness of the educational experience. I aim to develop frameworks and systems for informational dialogues with virtual agents as well as for emotional empathy and personalized interactions.

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